Narcissus Misgrounding: When Models Answer Without the Fact a Reform Made Decisive

Daniel GUEZ [a,1] and Rose CYMBLER [a]
[a] *Talia, Paris, France*

Abstract. We identify and measure narcissus misgrounding: language models answer a legal question without the fact that decides it, having supplied that fact itself. Nothing is fabricated, the law cited is correct, and the applicable version has been retrieved; what is missing is a fact about the case, which no corpus contains. The models know how to qualify an answer, but not on what is missing.

We build a benchmark of thirteen items across six reforms of French tax law. Each item is a file put to the model as a client would put it to counsel, paired with the version of the text applicable at the date of the facts, so that retrieval is not what is under test. Six items carry a fact the reform has made necessary and the file does not state; seven control items carry none. Thirteen models, three repetitions, 500 responses across 507 cells.

Across five of the six measurement items, those built from reported cases, models request the missing fact in only 3% of responses (5 of 192); on three of them the fact is never requested, and nine of the thirteen models never ask for it at all. The single outlier driving this figure to 16% is an AI-generated item, which differs from the other five on two dimensions at once, as Section 6 discusses. They do not ask indiscriminately either: on control items, where nothing is missing, they answer rather than ask in 96.5% of cases, apart from one item where they ask the question a later version of the text would call for.

Read by area, the request rate is the same on both sides of each reform: what models ask follows the subject of the file. A second author verified the coding on 135 cells drawn blind, including the five positive cases. All data and code are released.

Keywords. narcissus misgrounding, information elicitation, legislative reform, French tax law, benchmark

1. Introduction

A tax reform can change not only the answer to a question but also the facts needed to answer it.

Article 150-0 B ter of the French tax code allows a company director to defer tax on a capital gain when contributing shares to a holding company. If the holding company subsequently sells those shares within three years, it must reinvest most of the proceeds

[1] Corresponding author: Daniel Guez, Talia, Paris, France.

in an active economic business. Until 21 February 2026, property dealing (marchand de biens) qualified as an eligible activity. From that date, it is excluded.

Consider a seasoned director who has executed such restructurings for years. When consulting counsel after their holding company sells shares in February 2026 to reinvest in a property project, they have no reason to suspect a timing issue: under the law they know, the exact day of the sale is irrelevant. Yet their entire tax position hinges on that single unstated fact: before the 21st of February, the reinvestment is fully eligible; from the 21st, tax deferral is forfeited. A human lawyer immediately asks for the exact date. The question arises from the reform and appears nowhere in the client's file.

We report what thirteen language models do when placed in counsel's position.

The phenomenon

We call it narcissus misgrounding: a model answers a legal question without requesting the fact that decides it, having silently supplied that fact itself. In the vocabulary of argumentation theory, the model covertly discharges a burden of establishment, supplying the very premise it should have questioned.

Narcissus is neither blind nor misinformed, and nothing in the water is hidden from him. Held spellbound by his own reflection, he never looks outward. A language model given the applicable statutory text is in that state: it looks at what it already holds rather than at the client's case. What it lacks is a factual premise it could obtain by asking. Instead it fills the void from the default case, the ordinary practice, the scenario its training data suggests. The gap closes without a trace.

This makes the failure more dangerous than hallucination, which a reader can catch. A narcissus-misgrounded answer appears pristine: correct law, right version, seamless reasoning, resting on an assumed fact that silently invalidates the advice.

Contributions

Conceptualisation: we identify and characterize narcissus misgrounding as a structural failure mode distinct from retrieval failure and hallucination. *Benchmark*: we build thirteen legal files across six French tax reforms, each paired with the statutory text in force at the date of the facts, and evaluate thirteen leading models over three repetitions; all 573 responses (500 from the main campaign and 73 from the prompt ablation reported in Section 5.5), annotations, and code are released.[2] *Findings*: on the five reported-case items, models request the decisive fact in 3% of responses; the overall 16% rate is driven by one AI-generated item, which differs from the other five on two dimensions at once, as Section 6 discusses. *Resilience to prompting*: the failure survives an explicit instruction to ask (Section 5.5). *Observations on the instruments*: unenforced token ceilings, non-termination, and routing variability.

Why this is not a retrieval problem

Temporal Misgrounding in Legal RAG [1] demonstrated that a versioned corpus with date-conditioned retrieval recovers the applicable statutory version where standard retrieval defaults to the current text. Every prompt in this study directly provides the ap-

[2] Available at `https://github.com/DanielGuez16/narcissus-misgrounding`.

plicable legal text. The model holds the exact law in front of it. What it lacks is a client-specific fact, one that resides in no legal database or corpus, but must be requested.

2. Related work

Asking for what is missing. Steenhuis & Westermann [2] test whether an LLM requests missing information during legal aid intake; their most frequent error is asking unnecessary questions where a file could be decided. Two elements differ: their prompt explicitly commands asking, and their missing facts are statically enumerated by eligibility rules. Their work concerns guided intake, whereas we evaluate open advice. Steenhuis [3] extends this to triage, and Steenhuis and Harvey [4] extend it further, evaluating the quality of the follow-up questions themselves. Yu et al. [5] study interactive legal consultations, and Zhang et al. [6] evaluate legal consultation over several turns, where the exchange with the client is the object rather than a single answer. Outside law, QuestBench [7] and MediQ [8] test question-asking when required variables are fixed a priori. AskBench [9] measures the same behaviour outside law: models skip the clarification even where asking would improve the answer. The abstention literature measures whether models decline unanswerable questions [10], while Guitton et al. [11] measure textual ambiguity in rules. In our benchmark, the required variable is fixed only after a legislative reform alters the legal threshold.

Argumentation theory. Argumentation schemes carry critical questions [12] required before accepting conclusions. Carneades formalizes these as premises carrying specific burdens of proof [13]. Applying a scheme without raising its critical question leaves it dependent on an unstated assumption. Furthermore, Walton & Krabbe [14] distinguish information-seeking dialogue from persuasion dialogue; the evaluated models bypass the former entirely.

Norm change and the temporal dimension of law. A legal change can alter the conditions of a rule as well as its outcome. Governatori & Rotolo [15] separate abrogation from annulment in defeasible logic. Tamargo et al. [16] formalize temporalized belief revision, extending Lindahl & Odelstad [17] and Moguillansky et al. [18]. Palmirani et al. [19,20] and Szőke et al. [21] model legal temporal dimensions and change management. Slingerland et al. [22] measure reform impact on case classification, and de Martim [23] proposes point-in-time reconstruction for legal AI. We measure how models behave when one of these newly required conditions goes missing.

Reliability of language models in law. Studies document persistent legal AI fragilities: uncorrected user premises [24], unsupported RAG citations [25], and response instability at zero temperature [26]. Models fail on explicit statutory rules [27], subsumption [28], and detecting overruled precedent [29], while Gultekin et al. [30] decouple factual accuracy from reasoning quality. LegalBench [31] unifies such tasks. Crucially, these benchmarks evaluate answer correctness or abstention; we measure an upstream failure: whether the model registers that it cannot answer without eliciting missing facts. Lou and Shin [32] make a related point: legal benchmarks feed models clean expert inputs, and hide the gap with what a layperson actually brings.

What has not been measured. Across the venues and period surveyed, we find no study evaluating whether a legal AI system actively requests factual information that a legislative reform has newly made decisive.[3]

3. The case

A company director sells their business in two steps: they contribute their shares to a holding company, and the holding company subsequently sells them.

The shares cost the director one million euros, contributed at an agreed valuation of nine; less than three years later, the holding company sells them for nine million euros.

The contribution, and why tax is deferred. For tax purposes, a contribution of shares is treated as a sale. The eight-million-euro capital gain is in principle taxable even though the director received no cash, creating an immediate tax liability of 2.4 million euros without liquid funds to pay it. Article 150-0 B ter of the French Tax Code defers this tax liability.

The disposal, and the obligation it creates. The holding company no longer owns a business; it holds nine million euros in cash. The director has personally sold nothing, retaining their shares in the holding company. Because the disposal occurred within three years of the initial contribution, the statute requires the holding company to reinvest a substantial portion of the sale proceeds into an active economic business. How much must be reinvested, and within what timeframe, are conditions directly altered by the reform: 60% within two years for disposals prior to 21 February 2026, versus 70% within three years for disposals from that date onward. That represents a choice between placing 5.4 million euros within two years or 6.3 million euros within three. The reform also raises the minimum holding period on what is acquired, from twelve months to five years.

Where the lawyer stops and the system proceeds. The director consults their tax counsel:

> *I contributed my shares to my holding company, and it sold them in February 2026. I want to reinvest in a company that buys buildings, renovates them, and resells them. Does that qualify?*

The client states everything they believe to be relevant. One decisive fact is missing, omitted precisely because they have no reason to know it matters: on which exact day in February did the holding company complete the sale?

Article 11 of the amending act provides, in an uncodified transitional clause, that the reform applies to disposals executed from the day after publication. The law was published on 20 February; the legal boundary falls strictly on the 21st. Prior to the 21st, the eligible reinvestment scope includes any commercial activity under Articles 34 and 35 of the code, under which property dealing (marchand de biens) qualifies. From the 21st onward, the text references a provision excluding real-estate activities. After that date, the project fails.

[3] We searched the full proceedings of JURIX 2023 to 2025, ICAIL 2023, 2025 and 2026 (including its workshops), and the journal Artificial Intelligence and Law since 2023.

A system with a versioned corpus would find both versions and have to choose between them.

The human tax lawyer, by contrast, picks up the telephone. Neither the lawyer nor the model can conclude without the date, but the lawyer asks for it, recognizing that a single missing day dictates four terms of the opinion: project eligibility, the reinvestment quota, the compliance deadline, and the minimum holding period, twelve months before the reform and five years after.

4. Experimental design

We set out to measure errors that a reader of the answer cannot see.

The task. Each item is a tax file presented to the model as a client would submit it to counsel, accompanied by the statutory text in force at the date of the facts. The system prompt is minimal and unguided across all runs:

You are a tax lawyer. A client is consulting you. Answer as you would in a real file.

The items, the statutory texts and the system prompt are in French; the prompt is translated here. A complete item, with its statutory text, is in the repository.

The prompt does not ask the model to look for missing facts. Counsel receiving a file is not warned that a fact is absent, and such an instruction would give the test away.

Retrieval is assumed solved, for the reason given in Section 1.

4.1. The items

Table 1. The six areas of French tax law: the effective date of each reform, and the enacting text with its date of publication in the Journal officiel.

Area	Effective	Enacting law (JO)
Reduced VAT on renovation	1 Mar 2025	Loi 2025-127, JO 15 Feb 2025
Registration duties on property-holding companies	1 Jan 2024	Loi 2023-1322, JO 30 Dec 2023
Partial contribution of assets	1 Jul 2023	Ord. 2023-393, JO 25 May 2023; CGI aligned by Loi 2025-127
Withholding tax and beneficial ownership	16 Feb 2025	Loi 2025-127, JO 15 Feb 2025
Dutreil business transfer relief	17 Oct 2023	Loi 2023-1322, JO 30 Dec 2023
Contribution-and-disposal relief	21 Feb 2026	Loi 2026-103, JO 20 Feb 2026

Twelve items pair each of the six areas before and after its reform. The contribution-and-disposal area carries a thirteenth, dated February 2026 without the day, which straddles the reform.

On a control item nothing is missing and the model should answer; the controls show whether a request rate means anything.

Table 2. The thirteen items. Condition: the date of the facts relative to the reform.

Item	Area	Condition	Role	Missing fact
f001_chaudiere_a	Reduced VAT on renovation	Before	Control	–
f001_chaudiere_b	Reduced VAT on renovation	After	Measurement	The fuel the boiler can burn
f002_preponderance_a	Property-holding companies	Before	Control	–
f002_preponderance_b	Property-holding companies	After	Measurement	Three declarations in the deed
f003_apport_partiel_a	Partial contribution	Before	Control	–
f003_apport_partiel_b	Partial contribution	After	Measurement	The date the plan was filed with the registry
f004_beneficiaire_effectif_a	Beneficial ownership	Before	Control	–
f004_beneficiaire_effectif_b	Beneficial ownership	After	Measurement	What the parent does with the funds
f005_dutreil_a	Dutreil	Before	Control	–
f005_dutreil_b	Dutreil	After	Measurement	The services provided to the occupants
f006_apport_cession_a	Contribution-and-disposal	Before	Control	–
f006_apport_cession_b	Contribution-and-disposal	After	Control	– (day and sector given)
f006_apport_cession_c	Contribution-and-disposal	Straddling	Measurement	The day of the disposal, and the sector of the reinvestment

4.2. Models and runs

The benchmark evaluates thirteen leading language models, queried between 05:15 and 14:07 UTC on 19 August 2026. Claude Opus 5 was reached through Anthropic's first-party API; GPT-5.6 Sol, GPT-5.6 Terra, Gemini 3.1 Pro, Grok 4.6, Mistral Large 3, Mistral Medium 3.5, and Qwen 3.6 Plus through OpenRouter to their first-party backends; and Kimi K3, GLM-5.2, DeepSeek V4 Pro, MiniMax M3, and Llama 4 Maverick across third-party providers. The last eight have published weights, though all runs used cloud endpoints. Complete identifiers, routing metadata, and timestamps are in the released dataset. Temperature was zero wherever the endpoint supports it; Claude Opus 5 and the two GPT-5.6 variants do not expose the parameter. Three repetitions per item yield 500 responses across 507 cells, after seven API non-terminations. Of the four models that produce the five requests on real-world items, two were routed to a single backend and two across several: the figure is not an artefact of routing.

4.3. Coding

Each response is placed in one of four categories: it requests the missing fact and states no conclusion; it answers, then requests the fact as a condition; it assumes a value and concludes on it; or it does not ask and concludes with no mention that anything is missing. A fifth code, *absent*, marks seven cells where a model produced no response, excluded from every denominator. On a control item nothing is missing, so only two codes

can apply, the model answers or requests something it does not need; this held across all 269 coded control cells.

Automatic annotation was performed by Claude Opus 5. *Temporal Misgrounding in Legal RAG* [1] rejected LLM-as-a-judge because a judge scoring legal accuracy would inherit the temporal biases it was meant to detect. That objection does not apply here: our evaluator assesses no legal reasoning, only whether the model explicitly asked for the identified missing fact, receiving the case, the missing fact and the output with all model identifiers stripped. To prevent annotation hallucination, it must quote verbatim the sentence justifying its code, and an automated check rejects any output where that quote is not an exact substring of the response. Although Claude Opus 5 is itself one of the thirteen evaluated models, double-blind human verification found 100% agreement ($n = 13$) on the responses it produced, against 98% ($n = 122$) elsewhere, ruling out self-evaluation bias.

4.4. Human verification

A second author independently coded 135 cells in a double-blind setup, unaware of the automated annotations, model identities, and item labels. Sampling was conducted across three draws: 50 uniform cells to mirror dataset distribution; 40 stratified cells (10 per category) to test category separability; and a targeted draw of 45 cells focused on practitioner-sourced items.

Table 3. Inter-coder agreement between the automatic coding and a second author, on three blind draws.

Draw	n	Agreement	κ (4 codes)	Agreement	κ (reported variable)
Uniform	50	86.0%	0.776	98.0%	**0.922**
Stratified	40	82.5%	0.767	97.5%	**0.935**
Targeted	45	80.0%	0.704	100%	**1.000**

The reported rates rest on the binary variable, request or not, which a disagreement between *assumes* and *does not ask* leaves unchanged; the binary kappa ($\kappa \geq 0.922$) is the one that bears on Section 5.

Positive cases are rare on practitioner-sourced items (5 in 192 cells), so the third draw took all five with 40 negative cells. Both coders agreed on all five ($\kappa = 1.00$ over the 45 cells); the disagreements fall between *assumes* and *does not ask* (four-code $\kappa = 0.704$), which affects only the discussion in Section 6.

5. Results

5.1. How often they ask

Across the five measurement items built from reported court cases, models request the missing fact in only 3% of responses (5 of 192). In the remaining 97%, they deliver a legal conclusion.

Table 4. Behaviour across the five practitioner-sourced measurement items (192 responses).

Behaviour	n	%
Requests the missing fact and withholds a conclusion	5	**2.6**
Answers, then requests the fact	37	19.3
Assumes a value for the fact and answers	75	39.1
Answers with no mention of the missing fact	75	39.1

Only the first behaviour is correct: legal counsel must ask before advising. The other three all deliver an actionable conclusion on an unverified premise, and we group them as misgrounding (187 of 192, 97.4%).

The overall request rate across all six items reaches 16% solely because of a single synthetic outlier: the AI-generated boiler item. Resampling models across the five real-world items yields a narrow confidence interval of [0.5%, 5.1%], whereas resampling items across the full set yields [0.4%, 44.4%]. This extreme width is a structural finding rather than a limitation: the variance resides entirely in the items.

A crossed random-effects logistic model over the 231 measurement cells places the item variance at 9.13 [3.34, 24.94] against 0.27 [0.12, 0.58] for the model. Fitted without the synthetic boiler item, the item variance drops to 1.22 [0.38, 3.94] against 0.45 [0.21, 0.96]. Across both fits, model variance remains consistently narrow: which model is queried does not predict whether it will ask.

5.2. Model identity does not predict the behaviour

Four models request the missing fact at all: one twice, three once. The other nine never do. Each model produced fifteen responses on those five items, except MiniMax M3, which produced twelve.

The spread between the best and the worst model does not change the conclusion.

Repetition on the strongest item. Each item was put to each model three times. Across the corpus, two thirds of the model-item pairs return the same code on all three repetitions. On the registry-filing item the thirteen model-item pairs are unanimous, across thirty-eight responses: no model varies between its own repetitions.

Blair-Stanek & Van Durme [26] report instability of up to 50% at temperature zero; a third of our pairs vary, but not on this item.

Manual verification of the strongest figure. We read all thirty-eight conclusions on the registry-filing item rather than relying on the automatic coding for the figure that carries the paper. All thirty-eight decide the merits, several under reservation: *subject to the transferred branch qualifying as complete*, *subject to the three-year holding undertaking*. None of these reservations bears on the date that governs which regime applies.

5.3. Controls

Seven control items, on which nothing is missing, and the model should answer. It does so in 86.6% of cases, 233 of 269.

The remaining 13% are requests made where no fact is missing, and they concentrate on a single item. On that item the reform is not yet in force and the fuel type carries no consequence, yet twenty-eight of thirty-nine responses ask for it.

They ask the question the later version of the text would call for, on facts the earlier version governs. Excluding the boiler, the controls return eight false positives across 230 cells, or 96.5% correct.

5.4. What the models ask does not change across a reform

The requests follow the subject of the file rather than the timing of the reform.

Table 5. Request rate by area. The before column holds the controls, set before the reform; the after column holds the measurement item, except in the last row, where it holds the seventh control, and in the contribution-and-disposal row, where the measurement item straddles the reform.

Area	Before (%)	After (%)
Boiler	72 (28/39)	85 (33/39)
Property-holding companies	3 (1/38)	0 (0/39)
Partial contribution	5 (2/38)	0 (0/38)
Beneficial ownership	0 (0/39)	3 (1/38)
Dutreil	11 (4/37)	10 (4/39)
Contribution-and-disposal	0 (0/39)	0 (0/38)
Contribution-and-disposal (day and sector given)	–	3 (1/39)

Read across the rows rather than down the columns. On the boiler item the models ask on both sides of the reform, although the fuel carries no consequence before it. On Dutreil they ask on both sides, at 11% and 10%. On the remaining four they ask on neither. The last row is a seventh control, set after the reform but giving the day and the sector: nothing is missing there either. It carries the eighth false positive.

The questions the models ask stay the same on both sides of a reform, so the new requirement in the law is not what prompts them.

The random-effects model puts more variance on the item than on the model, though on the five practitioner-sourced items alone the two are not separable. The subject of the file predicts whether a model will ask; the choice of model does not.

5.5. The failure survives explicit injunction

A reader might hypothesize that models fail to elicit facts simply because our baseline system prompt does not explicitly instruct them to do so. We empirically tested this hypothesis through an ablation study across two items, thirteen models, and three repetitions (78 calls, 5 API non-terminations). We appended a single explicit instruction to the system prompt:

If a necessary fact is missing, ask for it before concluding.

The two items diverge. On the Dutreil item, where the missing fact is factually salient, the request rate rises from 4 of 39 to 27 of 35. On the filing-date item, where the missing fact is a temporal condition created by the reform, it stays at 0 of 38.

Two conclusions follow. First, prompt engineering cannot resolve narcissus misgrounding when the factual void is induced by statutory change. Second, because the prompt in Section 5.4 was held constant on both sides of each reform, prompt wording cannot account for the cross-reform invariance we observe.

6. Why they do not ask

Each missing fact admits a default value the model can adopt silently. For *dates*, it defaults to the latest statutory text and applies it; for *occupant services* (Dutreil), it assumes a standard furnished lease without hotel services; for *statutory declarations*, it recites the list verbatim without registering that it lacks client values to fill it in; and for the *use of funds*, it answers under reservation, delivering a conclusion while treating the missing fact as a mere hypothetical.

When a default value is available, the model adopts it instead of asking, and the response never says that a value was chosen: this is what makes the error invisible.

The synthetic outlier. The boiler item stands apart because its missing fact admits no default, and it alone was AI-generated rather than drawn from a reported case. This may indicate that models pick up structural cues from their training distribution when faced with synthetic inputs, and default to silent assumptions on real-world files.

Observations on the instruments. Three deployment findings emerged. *Token ceilings*: MiniMax M3 exceeded its 32,000 token limit to return 37,016 reasoning tokens, and generated 79,371 at a 64,000 limit. *Non-termination*: the same model failed to conclude across seven cells, generating reasoning until hitting provider timeouts. *Routing variability*: four failed calls succeeded on identical retries where only the upstream API provider changed, revealing non-deterministic routing in aggregators.

7. Conclusion

Narcissus misgrounding sits downstream of the retrieval failure identified in Temporal Misgrounding in Legal RAG [1]. The model holds the right law at the right date; what it lacks is a fact about the client that no corpus contains. An opinion that cites the right law and reasons correctly can still rest on a factual assumption the reader cannot locate.

8. Limitations and Future Work

Five boundaries. *Domain scope*: six French tax regimes, though the mechanism, where a norm change creates an unstated factual requirement, is domain-agnostic. *Prompting regime*: an unguided baseline, with explicit elicitation tested by ablation on two items only. *Multi-fact interaction*: two items miss several facts at once; on the disposal item a date and a sector together induced four-code disagreement ($\kappa = 0.704$), leaving binary agreement unaffected ($\kappa = 1.00$). *Provenance*: one synthetic item against five drawn from reported cases; separating provenance from default availability needs real-world items with no default value. *Single-turn evaluation*: a single API call offers no channel for dialogue, and a model that answers is not necessarily blind to the gap. Multi-turn settings [6] would separate the two.

Declaration on the use of AI systems. The thirteen models are the object of the research, not an instrument of its writing: their responses are the data. The coding was performed by Claude Opus 5, declared and verified (Section 4.4). Code and writing used a language model, to the exclusion of content, references and results; the authors are solely responsible for every element presented.

References

[1] Cymbler R, Guez D, Fabre L. Temporal Misgrounding in Legal RAG: A Versioned-Corpus Benchmark for French Tax Law. In: ICML 2026 Workshop on AI for Law (AI4Law). Seoul; 2026. .

[2] Steenhuis Q, Westermann H. Getting in the Door: Streamlining Intake in Civil Legal Services with Large Language Models. In: Legal Knowledge and Information Systems (JURIX 2024). Frontiers in Artificial Intelligence and Applications; 2024. p. 155-67. doi:10.3233/FAIA241242.

[3] Steenhuis Q. That's So FETCH: Fashioning Ensemble Techniques for LLM Classification in Civil Legal Intake and Referral. In: Legal Knowledge and Information Systems (JURIX 2025). Frontiers in Artificial Intelligence and Applications; 2025. p. 192-203. doi:10.3233/FAIA251588.

[4] Steenhuis Q, Harvey J. On Wednesdays, We Ask Questions: Optimizing "Active Listening" in Automated Legal Triage and Referral. In: AIDA2J Workshop at the International Conference on Artificial Intelligence and Law (ICAIL 2026). Singapore; 2026. .

[5] Yu YC, Huang SC, Shao HL. A Multi-Agent AI Framework for Legal Aid: Real-World Interactive Dialogue in Case Consultations. In: Legal Knowledge and Information Systems (JURIX 2025). Frontiers in Artificial Intelligence and Applications; 2025. p. 228-38. doi:10.3233/FAIA251591.

[6] Zhang L, Shi Y, Hu Y, Zhang J, Lv W, Ma Y, et al.. DLawBench: Evaluating LLMs Through Multi-Turn Legal Consultation; 2026. arXiv:2606.13931.

[7] Li BZ, Kim B, Wang Z. QuestBench: Can LLMs Ask the Right Question to Acquire Information in Reasoning Tasks?; 2025. arXiv:2503.22674. doi:10.48550/arXiv.2503.22674.

[8] Li SS, Balachandran V, Feng S, Ilgen JS, Pierson E, Koh PW, et al. MediQ: Question-Asking LLMs and a Benchmark for Reliable Interactive Clinical Reasoning. In: Advances in Neural Information Processing Systems (NeurIPS 2024); 2024. .

[9] Zhao J, Fang K, Cheng L. When and What to Ask: AskBench and Rubric-Guided RLVR for LLM Clarification; 2026. arXiv:2602.11199.

[10] Rajpurkar P, Jia R, Liang P. Know What You Don't Know: Unanswerable Questions for SQuAD. In: Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL 2018); 2018. p. 784-9. doi:10.18653/v1/P18-2124.

[11] Guitton C, Gubelmann R, Karray G, Mayer S, Tamò-Larrieux A. Identifying Open-Texture in Regulations Using LLMs. Artificial Intelligence and Law. 2025. doi:10.1007/s10506-025-09450-0.

[12] Walton D, Reed C, Macagno F. Argumentation Schemes. Cambridge: Cambridge University Press; 2008.

[13] Gordon TF, Prakken H, Walton D. The Carneades Model of Argument and Burden of Proof. Artificial Intelligence. 2007;171(10-15):875-96. doi:10.1016/j.artint.2007.04.010.

[14] Walton D, Krabbe ECW. Commitment in Dialogue: Basic Concepts of Interpersonal Reasoning. Albany: SUNY Press; 1995.

[15] Governatori G, Rotolo A. Changing Legal Systems: Legal Abrogations and Annulments in Defeasible Logic. Logic Journal of the IGPL. 2010;18(1):157-94. doi:10.1093/jigpal/jzp075.

[16] Tamargo LH, Martínez DC, Rotolo A, Governatori G. An Axiomatic Characterization of Temporalised Belief Revision in the Law. Artificial Intelligence and Law. 2019;27(4):347-67. doi:10.1007/s10506-019-09241-4.

[17] Lindahl L, Odelstad J. Normative Systems and their Revision: An Algebraic Approach. Artificial Intelligence and Law. 2003;11(2–3):81-104. doi:10.1023/B:ARTI.0000046005.10529.47.

[18] Moguillansky MO, Martínez DC, Tamargo LH, Rotolo A. An Approach to Temporalised Legal Revision Through Addition of Literals. Artificial Intelligence and Law. 2024;32(3):621-66. doi:10.1007/s10506-023-09363-w.

[19] Palmirani M, Governatori G, Contissa G. Temporal Dimensions in Rules Modelling. In: Legal Knowledge and Information Systems (JURIX 2010). Frontiers in Artificial Intelligence and Applications; 2010. p. 159-62. doi:10.3233/978-1-60750-682-9-159.

[20] Palmirani M, Cervone L. Legal Change Management with a Native XML Repository. In: Legal Knowledge and Information Systems (JURIX 2009). Frontiers in Artificial Intelligence and Applications; 2009. p. 146-55. doi:10.3233/978-1-60750-082-7-146.

[21] Szőke Á, Förhécz A, Strausz G. A Unified Change Management of Regulations and their Formal Representations Based on the FRBR Framework and the Direct Method. In: Legal Knowledge and Information Systems (JURIX 2012). Frontiers in Artificial Intelligence and Applications; 2012. p. 147-56. doi:10.3233/978-1-61499-167-0-147.

[22] Slingerland R, Boer A, Winkels R. Analysing the Impact of Legal Change Through Case Classification. In: Legal Knowledge and Information Systems (JURIX 2018). Frontiers in Artificial Intelligence and Applications; 2018. p. 121-30. doi:10.3233/978-1-61499-935-5-121.

[23] de Martim H. An Ontology-Driven Graph RAG for Legal Norms: A Hierarchical, Temporal, and Deterministic Approach. In: Legal Knowledge and Information Systems (JURIX 2025). Frontiers in Artificial Intelligence and Applications; 2025. p. 282-7. doi:10.3233/FAIA251598.

[24] Dahl M, Magesh V, Suzgun M, Ho DE. Large Legal Fictions: Profiling Legal Hallucinations in Large Language Models. Journal of Legal Analysis. 2024;16(1):64-93. doi:10.1093/jla/laae003.

[25] Magesh V, Surani F, Dahl M, Suzgun M, Manning CD, Ho DE. Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools. Journal of Empirical Legal Studies. 2025.

[26] Blair-Stanek A, Van Durme B. LLMs Provide Unstable Answers to Legal Questions. In: Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025); 2025. p. 425-9. doi:10.1145/3769126.3769245.

[27] Blair-Stanek A, Holzenberger N, Van Durme B. Can GPT-3 Perform Statutory Reasoning? In: Proceedings of the Nineteenth International Conference on Artificial Intelligence and Law (ICAIL 2023); 2023. p. 22-31. doi:10.1145/3594536.3595163.

[28] Wardas O, Matthes F. LLMs for Legal Subsumption in German Employment Contracts. In: Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025); 2025. p. 169-76. doi:10.1145/3769126.3769209.

[29] Zhang L, Savelka J, Ashley KD. Do LLMs Truly "Understand" When a Precedent Is Overruled? In: Legal Knowledge and Information Systems (JURIX 2025). Frontiers in Artificial Intelligence and Applications; 2025. p. 239-50. doi:10.3233/FAIA251592.

[30] Gultekin S, Rossi Reich M, Galloni F, Lagioia F, Consiglio E, Sartor G, et al. A Two-Dimensional Evaluation Framework for Factual and Reasoning Assessment of LLMs in Legal Question Answering. In: Legal Knowledge and Information Systems (JURIX 2025). Frontiers in Artificial Intelligence and Applications; 2025. p. 86-97. doi:10.3233/FAIA251579.

[31] Guha N, Nyarko J, Ho DE, Ré C, Chilton A, Narayana A, et al. LegalBench: A Collaboratively Built Benchmark for Measuring Legal Reasoning in Large Language Models. In: Advances in Neural Information Processing Systems (NeurIPS 2023), Datasets and Benchmarks Track; 2023. .

[32] Lou A, Shin D. Legal Reasoning Is Not Lawyering: Rethinking Legal Benchmarks for Pro Se Access to Justice; 2026. arXiv:2606.23716.